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Software Requirements Specification (SRS)

Bangladesh SmartShield: Integrated Crime, Digital Safety & Consumer Protection System

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February 09, 2026

Abstract

Bangladesh SmartShield is a proposed integrated digital platform designed to enhance public safety, crime reporting, emergency response, mobile theft protection, and consumer rights enforcement in Bangladesh. The project aims to address the limitations of existing fragmented systems by combining multiple safety and protection services into a single, user-friendly application. SmartShield enables citizens to report crimes anonymously, request emergency assistance through real-time SOS and GPS tracking, report stolen mobile devices using IMEI numbers, submit consumer complaints under existing laws, and access crime and fraud heatmaps for better situational awareness. By promoting transparency, accessibility, and citizen participation, the system seeks to reduce underreporting of crime, improve emergency response efficiency, and strengthen digital governance. Designed in alignment with the Smart Bangladesh Vision 2041, Bangladesh SmartShield aspires to empower citizens, support law enforcement, and build public trust through technology-driven solutions.

Acknowledgements

We would like to express our sincere gratitude to our course instructor for providing valuable guidance, encouragement, and constructive feedback throughout the development of this project. Their insights and suggestions played a crucial role in shaping the concept and structure of Bangladesh SmartShield.

We are also thankful to our university for providing the necessary resources and academic environment that supported this work. Special appreciation goes to our classmates and peers for their cooperation, discussions, and support during the project development process.

Finally, we acknowledge the availability of online resources, research articles, and existing digital platforms that helped us understand real-world public safety challenges and inspired the design of this system.

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Chapter 1

Introduction

1.1 Purpose:

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Bangladesh SmartShield** system. SmartShield is designed to address existing challenges in crime reporting, emergency response, digital safety, mobile device theft, and consumer rights protection by providing a unified, technology-driven national platform. Many individuals hesitate to report crimes or consumer violations due to fear of identity exposure, social pressure, or lack of trust in existing processes; SmartShield aims to reduce these barriers through secure and anonymous reporting mechanisms.This SRS specifies the system's objectives, scope, user roles, and required functionalities, serving as a common reference for software developers, system analysts, project supervisors, and relevant authorities. The document also covers incentive-based mechanisms, such as consumer compensation in the form of a percentage-based reward for valid complaints, which are intended to encourage public participation and transparency. The SRS defines system behavior, constraints, and interactions without detailing implementation or design decisions, ensuring clarity and consistency throughout the system development life cycle.

1.2 Intended Audience:

This Software Requirements Specification (SRS) document is intended for the following stakeholders:

- **Software Developers** – to understand system requirements and implement system features correctly.
- **System Analysts and Designers** – to model the system architecture, workflows, and interactions among system components.
- **Project Supervisors and Faculty Members** – to evaluate the system design, feasibility, and overall project objectives.
- **Government Authorities** – to oversee national safety operations, policy implementation, and system governance.

- **Citizens of Bangladesh** – as primary users of the system for crime reporting, emergency alerts, mobile device protection, and consumer complaint submission.
- **Law Enforcement Agencies (Police)** – to receive, investigate, and manage crime reports using real-time data, evidence, and location-based information.
- **Directorate of National Consumer Rights Protection (DNCRP)** – to process consumer complaints, verify evidence, enforce penalties, and manage consumer compensation.
- **Mobile Network Operators** – to support IMEI-based mobile device blocking, SIM activation control, and stolen phone enforcement mechanisms.
- **Bangladesh Telecommunication Regulatory Commission (BTRC)** – to regulate IMEI databases, ensure legal compliance, and coordinate between mobile operators and government authorities.

This document will be used as a formal reference throughout the system design, development, testing, deployment, and evaluation phases of the Bangladesh SmartShield project.

1.3 Scope of the System

Bangladesh SmartShield is a centralized digital safety and protection platform designed to enhance public safety, emergency response, digital security, and consumer rights protection across Bangladesh. The system aims to integrate multiple safety-related services into a single platform to reduce reporting delays, improve transparency, and strengthen coordination between citizens and responsible authorities.

1.3.1 System Goals:

The primary goals of the SmartShield system are to:

- Provide a secure and accessible platform for crime reporting, including anonymous submissions.
- Enable quick emergency response through real-time SOS alerts and location sharing.
- Protect mobile devices by preventing the misuse and resale of stolen phones using IMEI-based control mechanisms.
- Ensure transparency and fairness in consumer rights enforcement through digital complaint management and compensation tracking.
- Support government agencies and law enforcement bodies with data-driven insights, crime analytics, and safety monitoring tools.

1.3.2 Major Functions:

The system provides the following major functionalities:

- **Crime Reporting and Case Tracking:** allows citizens to report crimes, upload evidence, automatically capture location data, and track case progress.
- **Emergency and Personal Safety Services:**includes SOS alerts, automatic location sharing during emergencies, and safety indicators for different areas.
- **Stolen Mobile Phone and IMEI Protection:**enables IMEI registration, stolen phone reporting, network-level device blocking, and last-known location tracking.
- **Consumer Rights Protection:**supports digital submission of consumer complaints, evidence verification, price checking, authority integration, and compensation mechanisms for valid claims.
- **Administrative and Analytics Dashboard:**provides authorized authorities with tools for case management, crime heatmaps, fraud detection, and reporting.

1.3.3 System Boundaries:

The scope of Bangladesh SmartShield is limited to providing digital reporting, coordination, monitoring, and analytical support. The system does not replace law enforcement agencies, judicial processes, or regulatory authorities. Physical investigations, legal actions, penalties, and enforcement remain the responsibility of the respective government bodies. SmartShield functions as a supporting platform that facilitates information flow, transparency, and timely response within existing legal and institutional frameworks.

Chapter 2

System Overview

2.1 Product Perspective

Bangladesh SmartShield is a centralized national digital platform designed to enhance public safety, emergency response, digital protection, and consumer rights enforcement. The system operates as an integrated solution that connects citizens with multiple external organizations, including law enforcement agencies, emergency services, consumer rights authorities (DNCRP), mobile network operators, and the Bangladesh Telecommunication Regulatory Commission (BTRC).

SmartShield functions as a coordination and information-sharing platform rather than a replacement for existing systems. It integrates with external services such as emergency hotlines (e.g., 999), IMEI databases, and authority-managed systems through secure interfaces. By acting as a unified digital layer, SmartShield improves efficiency, transparency, and response time across existing institutional frameworks.

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2.2 Product Functions

At a high level, Bangladesh SmartShield provides the following core functions:

- Secure and anonymous crime reporting with evidence upload and real-time case tracking.
- Emergency and personal safety services, including SOS alerts and live location sharing.
- Stolen mobile phone protection through IMEI registration, reporting, and coordination with mobile network operators.
- Digital consumer rights protection through complaint submission, evidence verification, and compensation tracking.
- Safety analytics, crime heatmaps, and fraud-prone area identification.
- Administrative dashboards for authorized authorities to monitor, manage, and analyze system data.

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2.3 User Classes and Characteristics

The system supports multiple user classes with different access levels and responsibilities:

- **Citizens**: Primary users who report crimes, request emergency assistance, register IMEI numbers, and submit consumer complaints. Users may have varying levels of technical literacy, requiring a simple and intuitive interface.
- **Law Enforcement Officers**: Authorized users who review crime reports, access evidence, update case statuses, and analyze crime patterns.
- **Consumer Rights Officials (DNCRP)**: Users responsible for managing consumer complaints, verifying evidence, enforcing penalties, and overseeing compensation processes.
- **Government Administrators**: Oversight users who monitor system performance, policy compliance, and national safety metrics.
- **Mobile Network Operator Personnel**: Technical users involved in IMEI-based device blocking and network-level enforcement.
- **BTRC Officials**: Regulatory users responsible for IMEI database management, compliance monitoring, and coordination with mobile operators.

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2.4 Operating Environment

The Bangladesh SmartShield system is expected to operate in the following environment:

- **Client Side**: Mobile applications (Android) and web-based applications accessible through standard web browsers.
- **Server Side**: Secure cloud-based or government-hosted servers.
- **Software**: Web frameworks, mobile operating systems, and relational database management systems.
- **Network**: Internet connectivity via mobile data networks or broadband connections.
- **Security**: Secure authentication, encrypted communication, and controlled access mechanisms.

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2.5 Design and Implementation Constraints

The design and implementation of Bangladesh SmartShield are subject to the following constraints:

- Compliance with national laws, regulations, and data privacy policies of Bangladesh.
- Dependence on coordination with government authorities, law enforcement agencies, DNCRP, mobile operators, and BTRC.
- IMEI-level enforcement must follow regulatory approval and technical limitations.
- The system must ensure confidentiality, integrity, and availability of sensitive user data.
- Development tools and technologies must align with academic and institutional guidelines.

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2.6 Assumptions and Dependencies

The successful operation of the SmartShield system depends on the following assumptions and dependencies:

- Citizens have access to smartphones or internet-enabled devices.
- Government agencies and regulatory bodies cooperate in system integration.
- Mobile network operators support IMEI-based device blocking mechanisms.
- Emergency services respond to alerts generated by the system.
- Stable internet connectivity is available in most operational areas.

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2.7 SDLC Approach

The Bangladesh SmartShield (BD SmartShield) project follows a structured seven-step Software Development Life Cycle (SDLC) approach to ensure systematic development, quality, and maintainability:

1. **Planning**
    - Defining project objectives, scope, and feasibility.
    - Identifying stakeholders, risks, and resource requirements.
2. **Requirement Analysis**
    - Gathering functional and non-functional requirements.

- Collecting requirements through documents, surveys, and stakeholder input.

3. **System Design**
   - Designing system architecture and data flow.
   - Preparing database schema and UML diagrams.
4. **Implementation**
   - Developing system modules based on approved designs.
   - Integrating frontend and backend components.
5. **Testing**
   - Performing functional, security, and performance testing.
   - Identifying and fixing defects.
6. **Deployment**
   - Deploying the system in the operational environment.
   - Configuring system settings and user access.
7. **Maintenance**
   - Monitoring system performance.
   - Providing updates and improvements based on feedback.

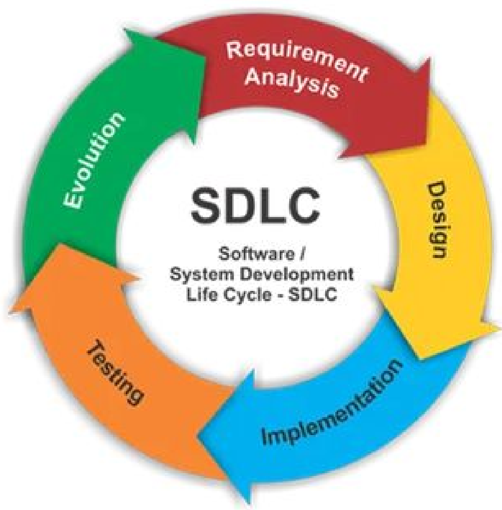

Figure 2.1: Phases of the SDL

Chapter 3

Requirements Elicitation & Analysis

This chapter describes the process used to identify, analyze, and finalize the requirements of the Bangladesh SmartShield system. It explains how information was collected, analyzed, and refined to ensure that the system addresses real-world problems and stakeholder needs.

3.1 Benchmark Analysis

Benchmark analysis was conducted to study existing systems and platforms related to public safety, emergency response, digital security, and consumer rights protection. The purpose of this analysis was to understand current solutions, identify best practices, and recognize limitations that SmartShield aims to overcome.

Examples of benchmark systems include:

- Emergency service applications (e.g., national emergency hotlines and SOS apps)
- Online crime reporting portals
- IMEI blocking and stolen phone tracking systems
- Consumer complaint management platforms
- Smart price checker

A comparative analysis was performed based on criteria such as feature availability, accessibility, transparency, response time, and system integration. The findings highlighted that existing systems are fragmented and lack a unified platform, which SmartShield addresses by integrating multiple safety and protection services into a single system.

Features	DNCRP Portal	Crime-Spotter	999 Site	Citizen COP	CEIR India	SmartShield (Our project)
Anonymous Reporting	N	Y	N	Y	N	Y
Evidence Upload	Y	Y	N	Y	N	Y
Consumer Complaint	Y	N	N	N	N	Y
Price/Product Fraud Reporting	Y	N	N	N	N	Y
SOS (Emergency + Geolocation)	N	Y	Y	Y	N	Y
IMEI Auto-Locking	N	N	N	N	Y	Y
Rewards	Y	N	N	N	N	Y
Auto Theft & Covert Capture	N	N	N	N	Y	Y
Safety Score / Risk Map	N	Y	N	N	N	Y
Smart Price Checker	N	N	N	N	N	Y

Table 3.1: Benchmark Analysis

3.2 Requirements Analysis

The purpose of requirements analysis is to identify and clearly define what the Bangladesh SmartShield system must do to meet stakeholder expectations. This process focuses on understanding user needs, system objectives, and operational constraints without considering implementation or technical design details.

The scope of requirement gathering includes:

- Functional requirements related to crime reporting, emergency alerts, IMEI protection, and consumer complaints
- Non-functional requirements such as security, usability, reliability, and performance
- Regulatory and legal requirements relevant to national safety and telecommunication systems

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3.3 Information Collected

During the elicitation process, various types of information were collected to ensure comprehensive requirement coverage:

- **Organizational Information**: Roles and responsibilities of government authorities, police, DNCRP, mobile operators, and regulatory bodies.
- **User Information**: Needs, expectations, and challenges faced by citizens when reporting crimes, seeking emergency help, or filing consumer complaints.
- **Process Information**: Existing workflows for crime reporting, emergency response, IMEI blocking, and consumer complaint handling.
- **Data Information**: Types of data involved, including personal information, location data, evidence files, IMEI numbers, and complaint records.

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3.4 Sources of Information

Requirements were gathered from multiple sources to ensure accuracy and relevance:

- **Internal Sources**: Project discussions, academic guidelines, and supervisor feedback.
- **External Sources**: Government portals, regulatory policies, and public service documentation.
- **Documents**: Existing system manuals, reports, and legal guidelines.
- **Observation**: Analysis of current processes used for crime reporting and consumer complaints.
- **Interviews**: Informal discussions with potential users and stakeholders.
- **Questionnaires**: Feedback collected from general users regarding safety concerns and usability expectations.
- **Online Resources**: Research articles, official websites, and case studies of similar systems.

3.4.1 Survey Result Table

Question	Decision
Q1: Which platforms have you used for crime-related information or help? • 56.7% Social Media (Facebook/WhatsApp) • 43.3% 999 Helpline • 33.3% I do not report incidents • 26.7% Police Station (Direct Visit) • 13.3% Inform Mobile App	Most people rely on social media for crime information. Official systems are used less; a unified platform is needed.
Q2: How often have you faced a situation where reporting a crime felt difficult or unsafe? • 24.1% Sometimes • 20.7% Rarely • 17.2% Often • 20.7% Very often • 17.2% Never	There is a strong need for anonymous and safe reporting.
Q3: What is the main reason people avoid reporting crimes, in your opinion? • 58.6% Fear of harassment • 48.3% Time-consuming process • 37.9% Do not know how to report • 48.3% Lack of trust • 55.2% No proper digital system	SmartShield must reduce fear, time complexity, and trust issues.

Table 3.2: Survey Result Table-1

Question	Decision
Q4: How familiar are you with any online crime-reporting platforms in Bangladesh? • 41.6% Not familiar • 19% Used once	SmartShield can become the first widely-used digital crime reporting system.
Q5: When reporting a crime, which part feels the most challenging? • 70.7% Providing crime details • 48.3% Uploading proof/evidence • 48.3% Fear of identity exposure	Need automated reporting flow with anonymity and status tracking.
Q6: Which consumer-related feature should existing apps offer but currently do not? • 37.9% Fraud history of shops • 37.9% Price comparison • 34.5% Product verification • 13.8% All of these	User's strongest demand is product verification (41.4%), followed by fraud history and instant price checking.
Q7: How would you rate your awareness of consumer rights in Bangladesh? • 41% Low • 34.5% Very low • 3.4% Very high	Price checker and fraud shop history can be strong unique USP features.

Table 3.3: Survey Result Table-2

<table>
<tr><th>Question</th><th>Decision</th></tr>
<tr><td>Q8: Where do you currently check product prices (food items, etc.)
41.4% Shop directly
34.5% Daraz
20.7% Friends/family suggestion
13.8% Others
3.4% d-araz</td><td>Focus on features that help users who do not usually check prices, such as smart price checking.</td></tr>
<tr><td>Q9: If you face overpriced goods or fraud, what action do you usually take?
57.1% Complain to the shop/staff
24.1% Leave the shop and ignore
14.3% Report to Consumer Rights (Vokta Odhikar)
14.3% Report online consumer rights groups
24.1% Do nothing</td><td>Prioritize easy, anonymous, one-tap reporting so users stop ignoring fraud.</td></tr>
<tr><td>Q10: Are you aware that under the Protection Act 2009, a consumer can receive 25% of the fine as a reward?
53.6% Yes, aware
20% No, I don't know
26.4% I am aware but don't know details</td><td>Include strong educational content to increase awareness of the 25% consumer reward system.</td></tr>
</table>

Table 3.4: Survey Result Table-3

<table>
<tr><th>Question</th><th>Decision</th></tr>
<tr><td>Q11: How aware are you of IMEI blocking systems for stolen phones?
22.3% I know how to block IMEI
40% I have heard about it
23.3% I don’t know how it works
13.3% I don’t know IMEI blocking exists</td><td>Add IMEI-blocking awareness tools and step-by-step guides to improve user knowledge.</td></tr>
<tr><td>Q12: Which crime-reporting features would you find most useful and safest?
70% Anonymous crime reporting
33.3% Uploading photos/videos
46.7% Automatic location tagging
26.7% Case status tracking
30% Nearby police station auto-routin</td><td>Prioritize anonymous crime reporting and automatic location tagging as core safety features.</td></tr>
<tr><td>Q13: How important is it that the system works even after a factory reset?
65.7% Very important
16.7% Important
10% Neutral
6.7% Not important</td><td>Ensure the system works even after factory reset as a top-level security requirement.</td></tr>
<tr><td>Q14: Which stolen-phone protection features do you find useful?
76.9% IMEI auto-lock
53.8% Factory-reset-resistant protection
28.5% Covert thief photo capture
34.2% Auto cloud backup evidence
4.2% Auto police report generation</td><td>Focus on IMEI auto-lock and factory-reset-resistant protection for stolen-phone security.</td></tr>
</table>

Table 3.5: Survey Result Table-4

Question	Decision
Q15: Which consumer-protection features would benefit you most? • 48.3% Smart price checker • 44.9% Fraud product reporting • 31% Complaint tracking • 55.2% 25% reward eligibility • 44.8% Nearby verified shops	Highlight reward eligibility, smart price checking, and verified shop features for consumer protection.
Q16: Which features make a digital platform trustworthy to you? • 51.7% Transparency • 51.7% Fast response • 48.3% Privacy protection • 48.3% Real-time updates • 44.8% Official authority integration	Strengthen platform trust through fast response, transparency, privacy, and authority integration.
Q17: Which features should be top priority for SmartShield? • 58.6% Crime reporting • 41.4% SOS emergency • 48.3% Safety alerts • 51.7% IMEI/stolen phone recovery • 48.3% Consumer rights reporting • 17.2% Price-checker • 17.2% Reward system • 3.4% Heatmap	Make crime reporting, IMEI recovery, safety alerts, SOS, and consumer-rights reporting top priorities for SmartShield.

Table 3.6: Survey Result Table-5

3.5 Gap Analysis

A gap analysis was performed to compare the current systems with the requirements of the proposed SmartShield system. Existing solutions were found to be limited by fragmentation, lack of transparency, delayed responses, and insufficient integration among authorities.

The proposed system addresses these gaps by:

- Providing a single unified platform instead of multiple isolated systems.
- Enabling anonymous reporting to reduce fear of identity exposure.
- Supporting real-time tracking and digital evidence submission.

- Integrating consumer complaint handling with compensation mechanisms.
- Offering data-driven analytics for better decision-making.

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3.6 Feature List Fixation

Based on the elicitation and analysis process, the final set of features supported by Bangladesh SmartShield has been finalized. These features include:

- Anonymous and identified crime reporting
- Evidence upload with location and time metadata
- Emergency SOS alerts with live location sharing
- IMEI registration and stolen mobile phone reporting
- Coordination with mobile operators for device blocking
- Digital consumer complaint submission and tracking
- Consumer compensation handling for valid complaints
- Administrative dashboards and analytics tools

These features represent the confirmed scope of the project and form the basis for detailed functional and non-functional requirements defined in subsequent chapters.

Chapter 4

Functional Requirements

This section describes the detailed functional requirements of the Bangladesh SmartShield system. Each requirement specifies system behavior, inputs, outputs, and interactions with users or external entities.

4.1 FR-1: User Registration and Authentication

Description: The system shall allow citizens and authorized personnel to register and authenticate securely in order to access SmartShield services.

Inputs:

- Full name
- National ID (NID) number
- Mobile number
- Email address
- Password

Outputs:

- Successful account creation confirmation
- Login access upon valid credentials
- Error message for invalid or duplicate data

Interactions:

- User interacts with the SmartShield web/mobile interface
- System securely stores encrypted credentials

4.2 FR-2: Anonymous Crime Reporting

Description: The system shall allow users to report crimes anonymously to reduce fear of identity exposure and encourage reporting.

Inputs:

- Crime type
- Description of the incident
- Optional evidence (photo, video, audio)
- Auto-detected location

Outputs:

- Unique case ID
- Case submission confirmation

Interactions:

- Police dashboard receives the report
- User identity remains hidden from law enforcement view

4.3 FR-3: Evidence Upload and Management

Description: The system shall allow users to upload digital evidence to support crime and consumer complaints.

Inputs:

- Image, video, or audio files

Outputs:

- Securely stored evidence with timestamp and location metadata

4.4 FR-4: Crime Case Tracking

Description: The system shall allow users to track the status of submitted crime reports.

Inputs:

- Case ID

Outputs:

- Case status (Received, In Review, Investigating, Resolved)

4.5 FR-5: SOS Emergency Alert

Description: The system shall provide a one-tap SOS emergency feature integrated with national emergency services.

Inputs:

- SOS button activation

Outputs:

- Live location sharing
- Emergency alert notification

Interactions:

- Sends alerts to police (999), trusted contacts, and nearby help centers

4.6 FR-6: IMEI Registration

Description: The system shall allow users to register mobile device IMEI numbers for theft protection.

Inputs:

- IMEI number
- Registered user NID

Outputs:

- IMEI registration confirmation

4.7 FR-7: Stolen Phone Reporting and IMEI Lock

Description: The system shall allow users to report stolen phones and automatically trigger IMEI-based locking.

Inputs:

- IMEI number
- Theft confirmation

Outputs:

- IMEI marked as stolen
- Device blocked from network access

Interactions:

- Integration with mobile operators and BTRC

4.8 FR-8: Factory Reset Resistant Tracking

Description: The system shall support tracking of stolen devices even after factory reset or SIM change.

Outputs:

- Last known device location

4.9 FR-9: Consumer Complaint Submission

Description: The system shall allow users to submit consumer rights complaints digitally.

Inputs:

- Complaint category
- Description
- Evidence
- Purchase details

Outputs:

- Complaint ID
- Submission confirmation

4.10 FR-10: DNCRP Complaint Processing

Description: The system shall forward consumer complaints to DNCRP for verification and enforcement.

Outputs:

- Complaint status updates
- Penalty and resolution notification

4.11 FR-11: Consumer Compensation Mechanism

Description: The system shall reward consumers with 25% of the penalty amount for valid complaints.

Outputs:

- Compensation notification
- Payment processing record

4.12 FR-12: Safety Map and Risk Zones

Description: The system shall display crime heatmaps and safety indicators.
Outputs:

- Red (high risk), Yellow (medium risk), Green (safe) zones

4.13 FR-13: Admin and Police Dashboard

Description: The system shall provide dashboards for police and administrators to manage cases.
Functions:

- View and update case status
- Analyze crime trends
- Monitor hotspot areas

4.14 FR-14: Notification and Alert System

Description: The system shall notify users about case updates, alerts, and safety warnings.
Outputs:

- Push notifications
- SMS or in-app alerts

Chapter 5

External & Non-Functional Requirements

This chapter describes the external interface requirements and non-functional requirements of the Bangladesh SmartShield system. These requirements define the quality attributes, system constraints, and interaction standards necessary to ensure reliability, security, scalability, and usability of the system.

5.1 User Interface Requirements

The Bangladesh SmartShield system shall provide a simple, intuitive, and accessible user interface suitable for citizens, police officials, and administrators.

5.1.1 Layout

- The system shall maintain a clean and consistent layout across all screens.
- Critical actions such as *SOS*, *Report Crime*, and *Submit Complaint* shall be prominently displayed.
- Color indicators shall be used to represent safety levels (Red = High Risk, Yellow = Medium Risk, Green = Safe).

5.1.2 Navigation

- The system shall provide menu-driven and intuitive navigation.
- Users shall be able to access core functionalities within three interactions.
- Navigation behavior shall remain consistent across mobile and web platforms.

5.1.3 Prototypes

- Interactive prototypes shall be developed during the design phase.
- User feedback shall be incorporated to improve usability and accessibility.

5.2 Software Interface Requirements

The system shall integrate with external software and government systems to ensure automated data exchange and service coordination.

- Integration with the national emergency service (999) for SOS alerts.
- Integration with Police Crime Management Systems.
- Integration with DNCRP systems for consumer complaint handling.
- Integration with BTRC databases for IMEI verification and device locking.
- RESTful APIs shall be used for secure communication between systems.

5.3 Performance Requirements

The system shall meet the following performance criteria to ensure smooth and efficient operation.

- The system shall respond to standard user requests within 2 seconds.
- SOS alerts shall be processed and transmitted within 1 second.
- The system shall support nationwide scalability with concurrent users.
- The system shall maintain high availability during peak usage periods.

5.4 Security Requirements

Security is a critical requirement due to the sensitive nature of personal, criminal, and emergency-related data.

- The system shall enforce secure authentication and authorization mechanisms.
- User data shall be encrypted during both transmission and storage.
- Role-based access control shall be implemented for Citizens, Police, and Administrators.
- Audit logs shall be maintained for all sensitive operations.
- The system shall prevent unauthorized access and data manipulation.

5.5 Maintainability

The system shall be designed to support ease of maintenance and future enhancements.

- The system architecture shall be modular and well-documented.
- New features shall be added without affecting existing system components.
- Maintenance and updates shall be deployable with minimal downtime.
- Standard coding conventions and documentation practices shall be followed.

5.6 Reliability and Fault Tolerance

The Bangladesh SmartShield system shall ensure continuous operation even under partial system failures.

- The system shall operate reliably under normal and peak load conditions.
- Automatic data backup and recovery mechanisms shall be implemented.
- In the event of network or server failure, the system shall recover without data loss.
- Critical services such as SOS alerts and crime reporting shall remain operational at all times.

Chapter 6

Feasibility & Project Management Analysis

This section evaluates the feasibility of developing and deploying the Bangladesh SmartShield system from multiple perspectives and analyzes its strategic viability using SWOT analysis.

6.1 Feasibility Analysis

6.1.1 Technical Feasibility

Bangladesh SmartShield is technically feasible as it relies on proven and widely used technologies such as web and mobile applications, cloud-based databases, GPS services, secure authentication mechanisms, and system integration APIs. Existing national infrastructures such as mobile networks, emergency services (999), and consumer rights authorities can be integrated through secure interfaces. The required technical skills and tools are available within the current software development ecosystem of Bangladesh.

6.1.2 Economic / Financial Feasibility

The system is economically feasible as it is designed as a centralized national platform that reduces duplication of services across multiple agencies. While initial development and integration costs may be moderate, long-term operational savings will result from automation, reduced manual processing, and faster resolution of cases. Additionally, improved crime prevention, theft reduction, and consumer protection can lead to indirect economic benefits for both the government and citizens.

6.1.3 Operational Feasibility

SmartShield is operationally feasible because it aligns with existing workflows of law enforcement agencies, emergency services, mobile operators, and consumer rights authorities. The system simplifies reporting, monitoring, and resolution processes, making it easier for authorities to adopt without major procedural changes. Citizens can operate the system using familiar smartphone and web interfaces, ensuring high usability.

6.1.4 Human Resource Feasibility

The project is feasible from a human resource perspective as it requires standard software development roles such as system analysts, developers, testers, and administrators. Government agencies already employ personnel capable of handling digital systems, and minimal training will be required for police officers and DNCRP staff to use the dashboards effectively.

6.1.5 Market, Legal, and Political Feasibility

The system is legally and politically feasible as it supports national priorities such as public safety, digital governance, and consumer rights protection. SmartShield complies with existing regulations related to telecommunications, consumer protection laws, and digital security policies. Strong government backing and public demand for safety and transparency further enhance its acceptance.

6.1.6 Schedule Feasibility

The project is schedule-feasible as it can be developed incrementally using a phased approach. Core features such as crime reporting and SOS alerts can be deployed first, followed by advanced modules like IMEI tracking and analytics. This staged development reduces risk and allows timely delivery within academic or institutional deadlines.

6.2 SWOT Analysis

6.2.1 Strengths

- Unified national platform combining multiple safety and protection services
- Encourages reporting through anonymous complaint mechanisms
- Integration with government authorities increases trust and credibility
- Consumer reward mechanism motivates valid complaints

6.2.2 Weaknesses

- High dependency on government and regulatory integration
- Requires strong data security and privacy management
- Initial adoption may be slow due to resistance to change

6.2.3 Opportunities

- Supports Digital Bangladesh and smart governance initiatives
- Can reduce crime, fraud, and mobile theft nationwide
- Potential future expansion to include AI-based crime prediction
- Increased public trust through transparency and accountability

6.2.4 Threats

- Cybersecurity threats and data breach risks
- Legal or policy changes affecting system operations
- Lack of coordination among multiple stakeholders
- Misuse of the system through false reporting

Chapter 7

System Design (Diagrams)

7.1 Data Flow Diagram (DFD)

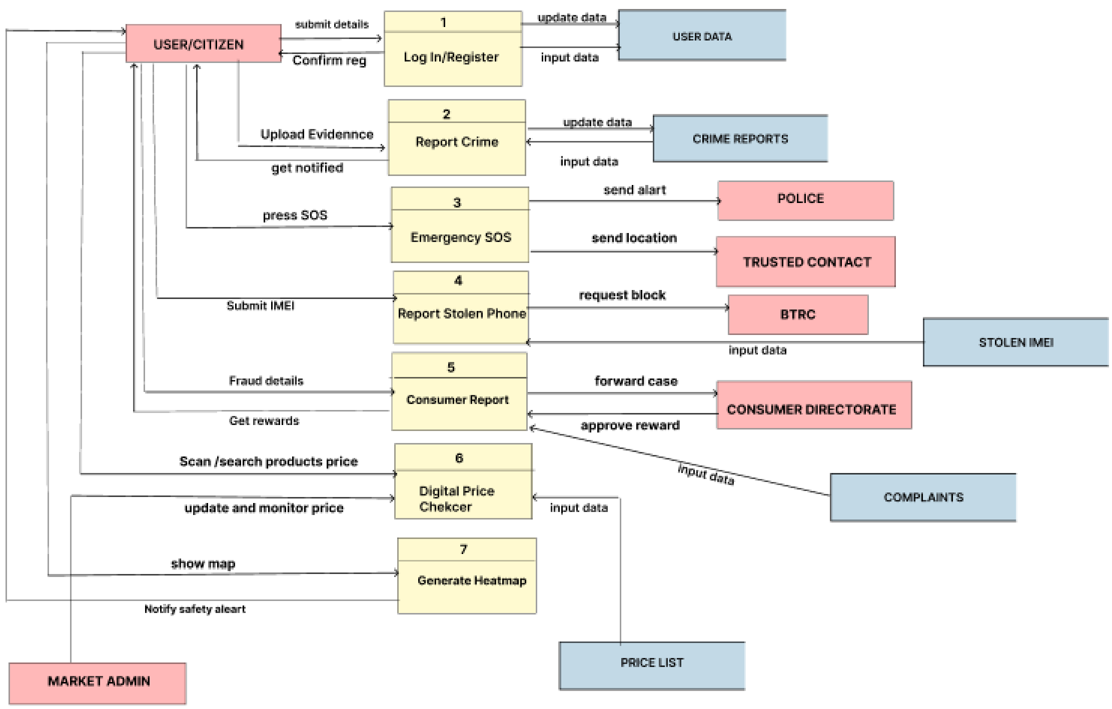

Figure 7.1: Data Flow Diagram of Bangladesh SmartSheild

7.2 Use Case Diagram

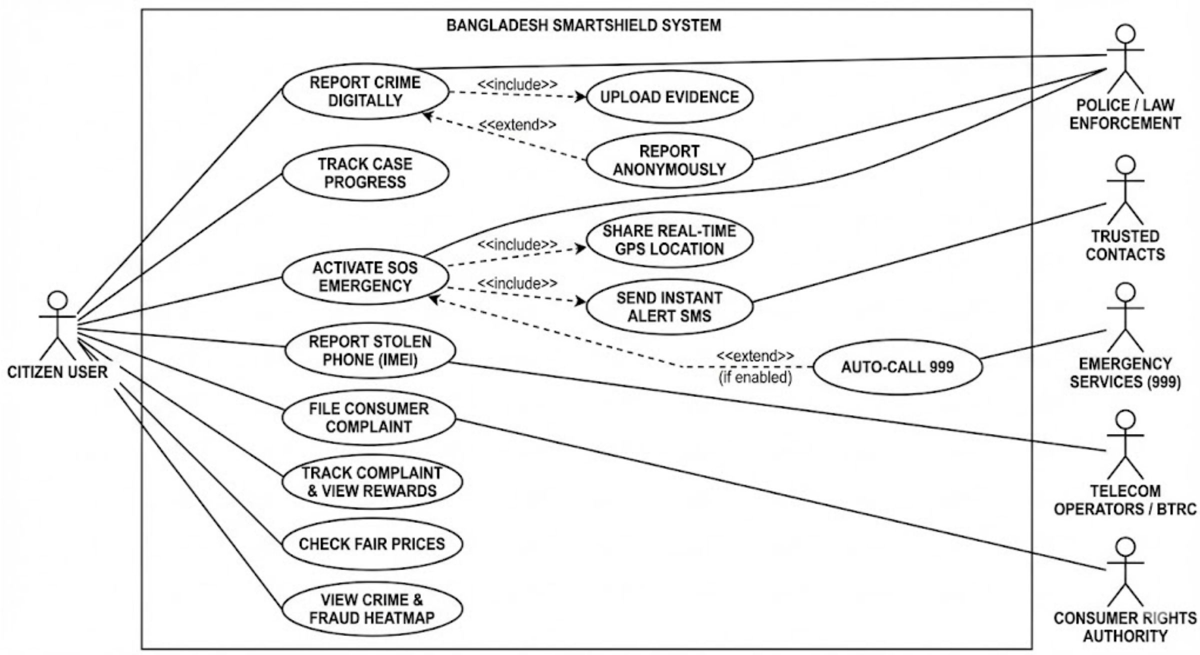

Figure 7.2: Use Case Diagram of Bangladesh SmartSheild

7.3 Class Diagram

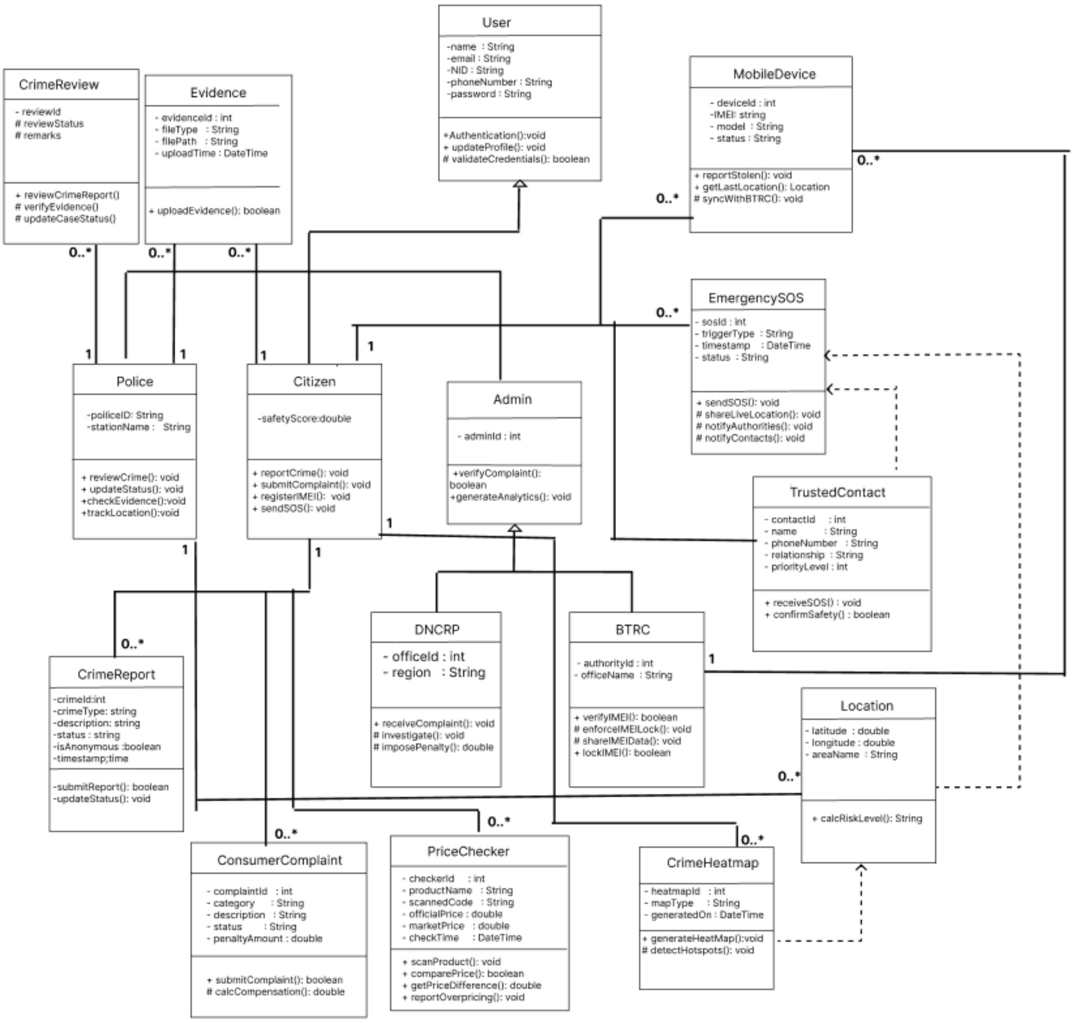

Figure 7.3: Class Diagram of Bangladesh SmartSheild

7.4 Sequence Diagram

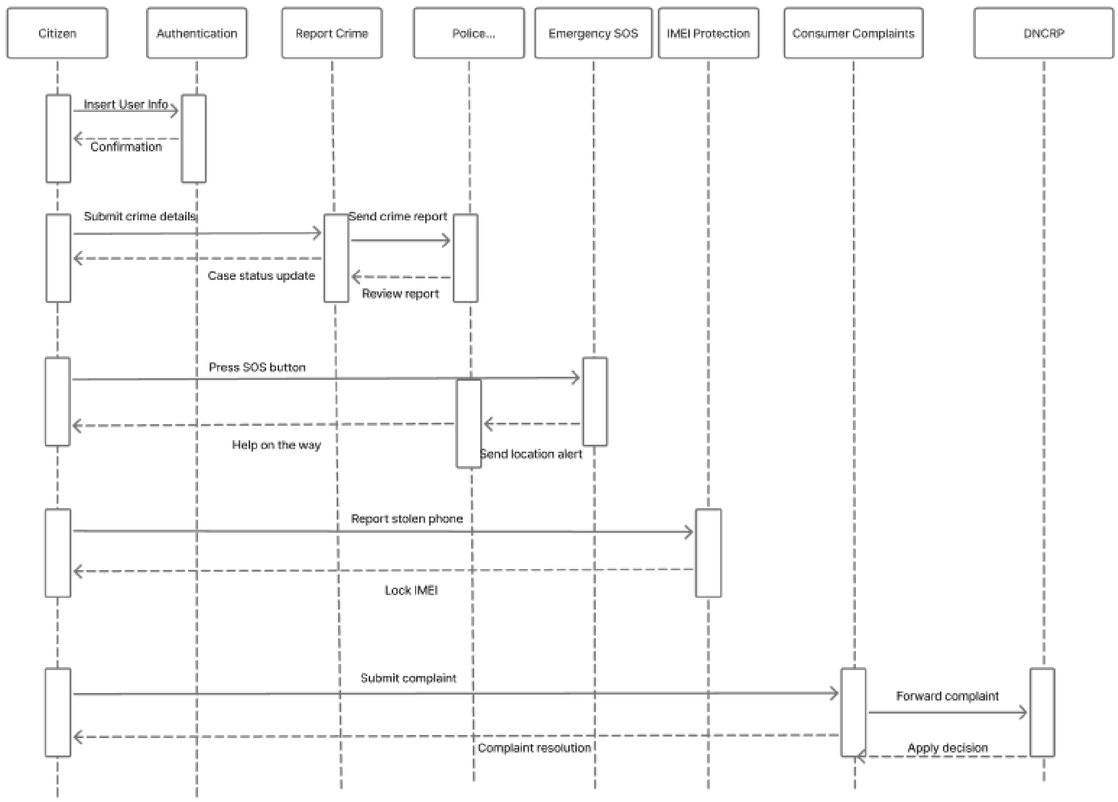

Figure 7.4: Sequence Diagram of Bangladesh SmartSheild

Chapter 8

Prototype Design

8.1 Screenshots of prototypes

8.1.1 Overview of Mobile Application

The user interface of the Bangladesh SmartShield mobile application has been designed to ensure a simple, intuitive, and smooth experience for all users. The design emphasizes clean layouts, easy navigation, and fully responsive components to ensure effective usability across different mobile devices.

Application Description

- **Authentication UI:** Supports login and registration.
- **Password Control:** Allows password visibility toggle.
- **Navigation Flow:** Ensures smooth screen transitions.

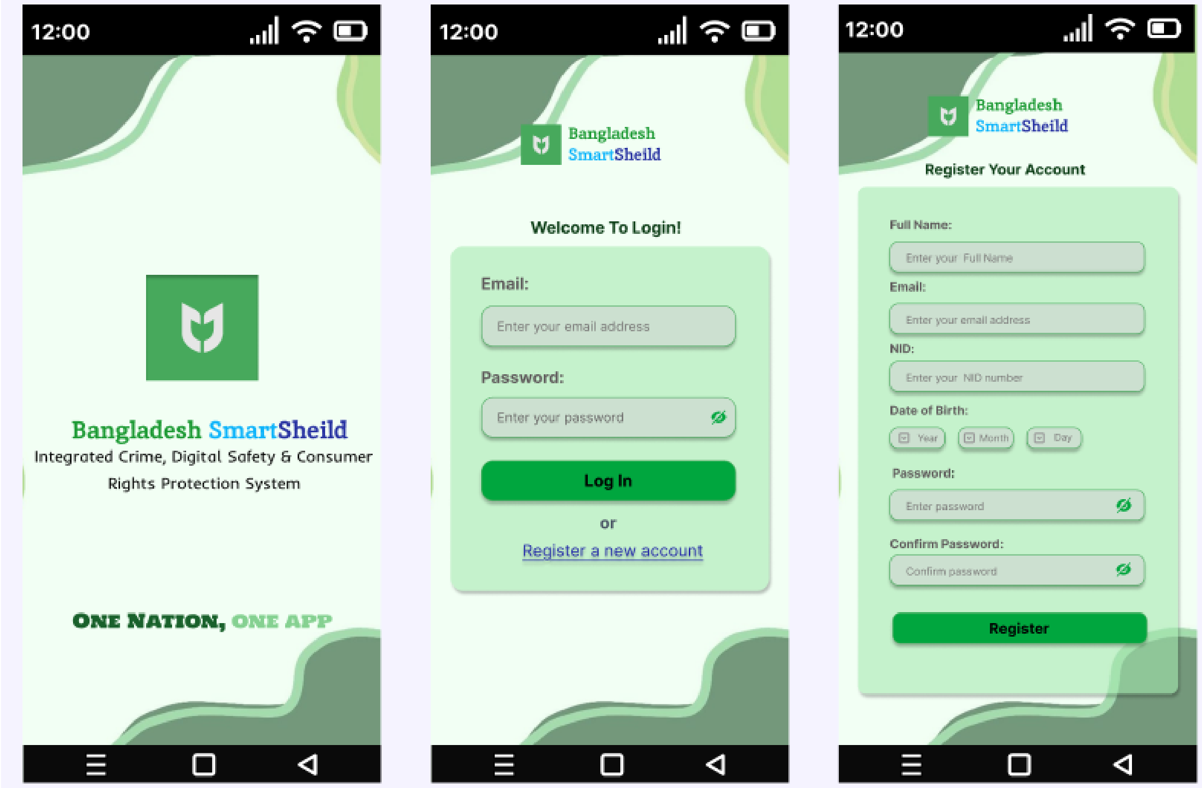

Figure 8.1: Homepage,Registration LogIn page

- **Dashboard:** Provides access to system features and summary data.
- **Core Actions:** Supports crime reporting, SOS, IMEI protection, and consumer rights.
- **Navigation:** Enables access to profile, analytics, and logout.
- **User Account:** Displays personal details and rewards.

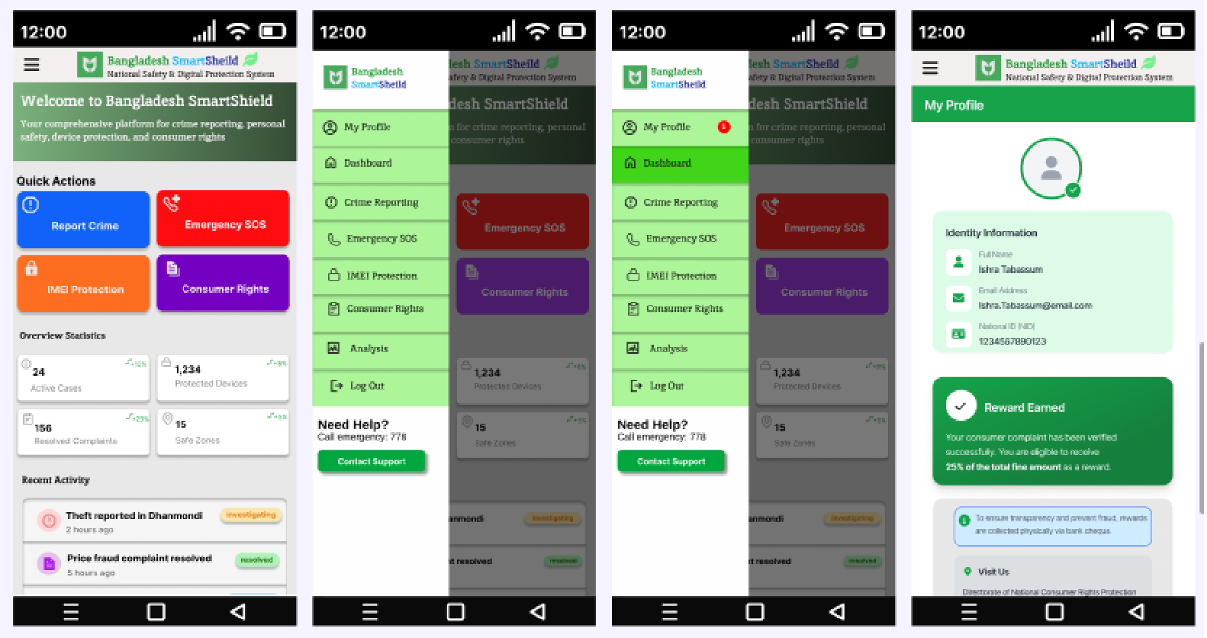

Figure 8.2: Dashboard Profile page

- **Crime Reporting:** Submit and track crime reports securely.
- **Anonymity & Evidence:** Supports anonymous reporting and evidence upload.
- **Case Status:** Displays report progress and investigation state.
- **Crime Heatmap:** Shows location-based crime risk levels.

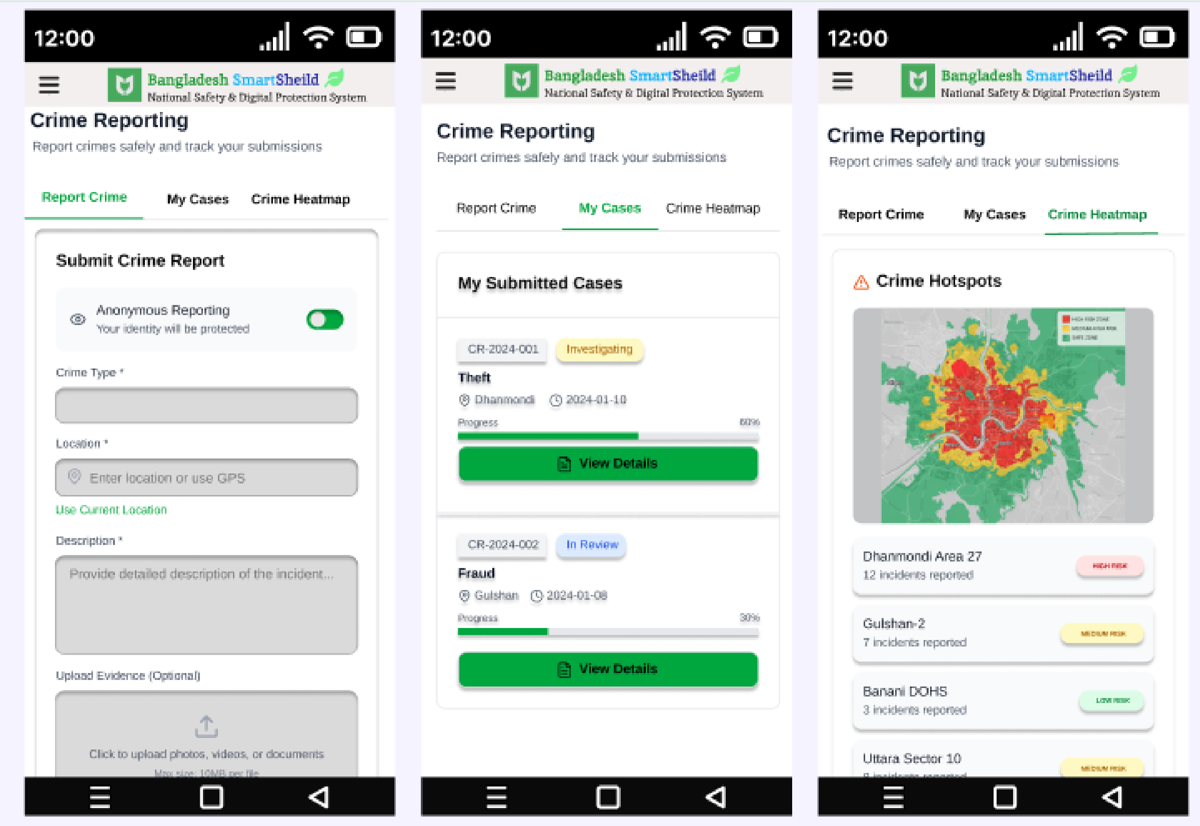

Figure 8.3: Crime Report pages

- **Emergency SOS:** Instantly contacts emergency services.
- **Location Sharing:** Sends real-time user location.
- **Trusted Alerts:** Notifies selected emergency contacts.

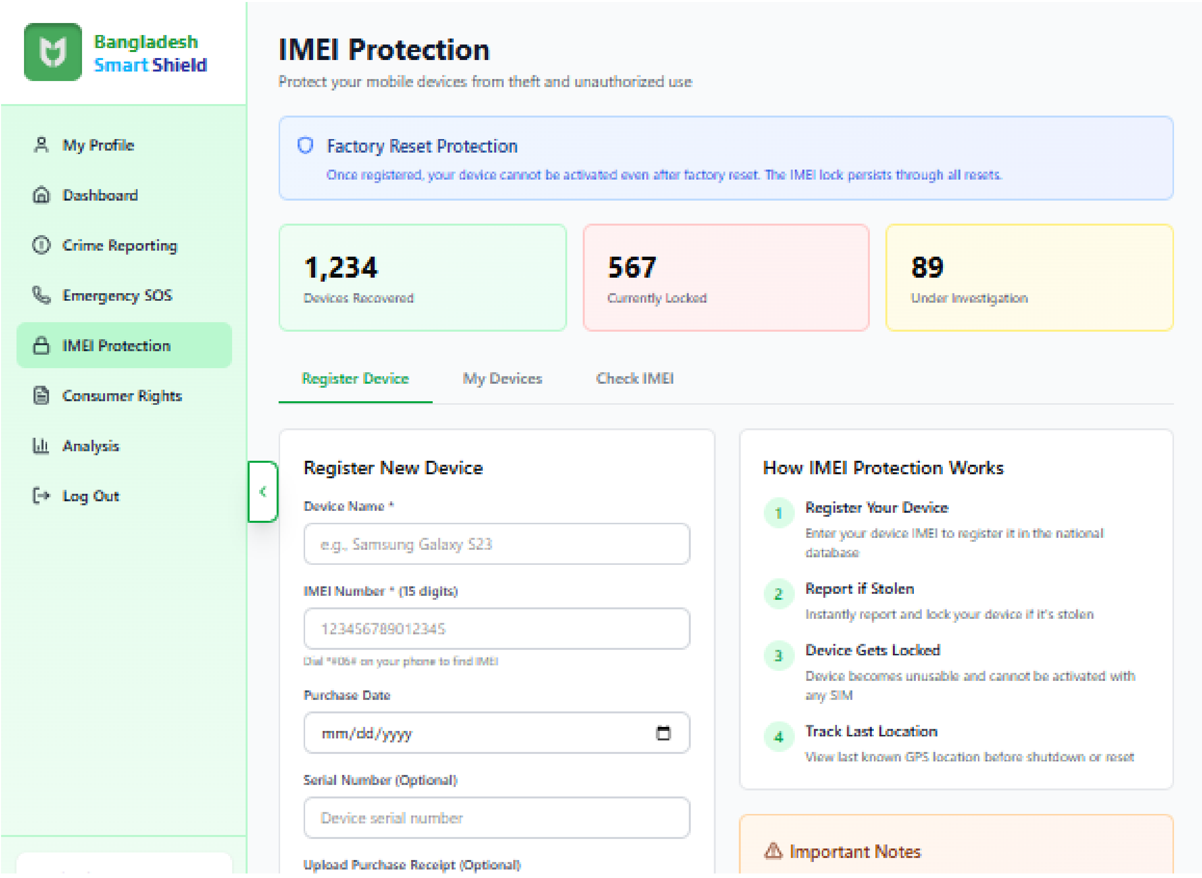

Figure 8.4: Emergency SOS Activation page

- **Consumer Complaints:** Submit and track consumer rights violations.
- **Price Verification:** Check product price and authenticity.
- **Compensation:** Displays rewards for verified complaints.

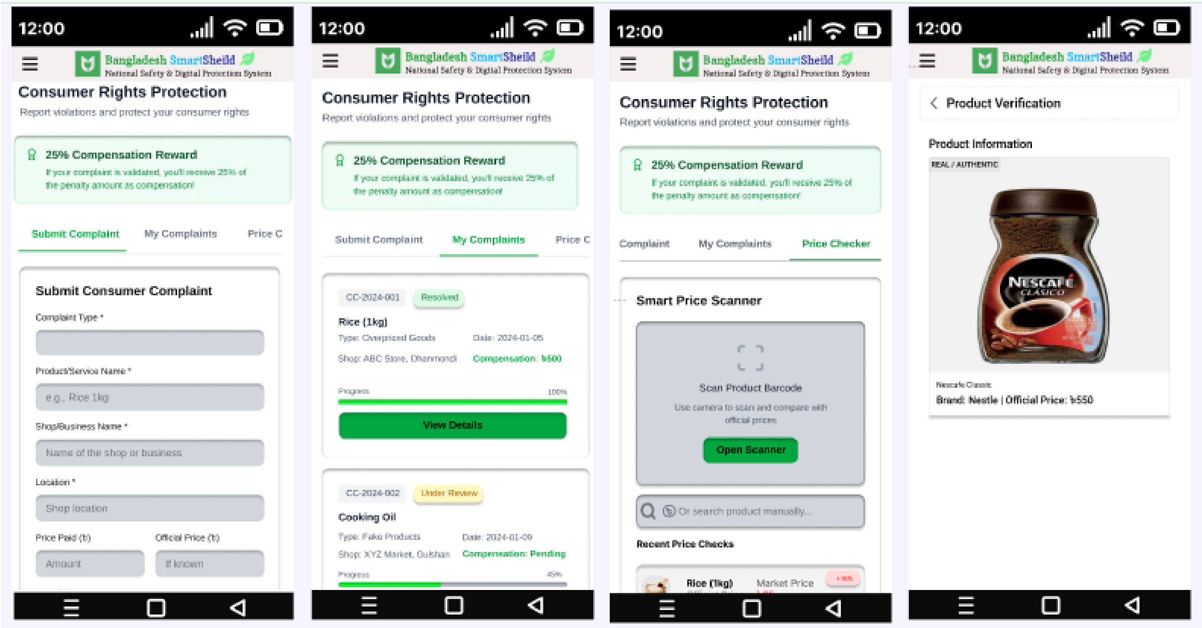

Figure 8.5: Consumer Rights Complaints smart price checker

- **IMEI Protection:** Registers and protects mobile devices.
- **Device Status:** Displays device tracking and recovery state.
- **IMEI Verification:** Checks IMEI validity before purchase.

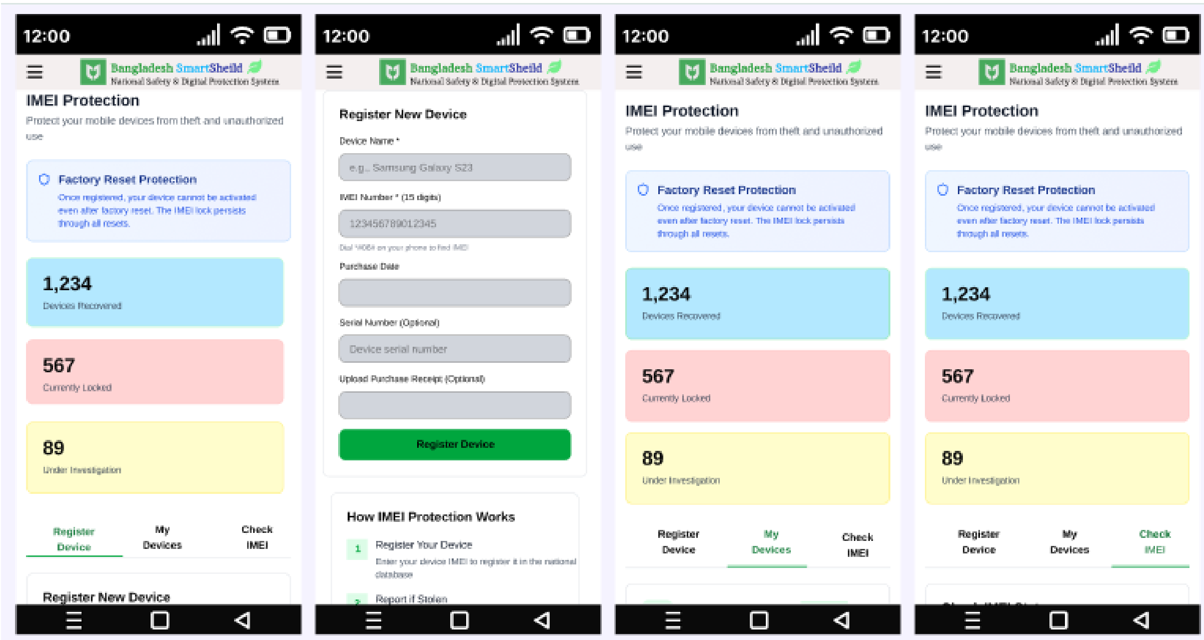

Figure 8.6: IMEI Registration pages

- **Analytics Panel:** Displays summarized system data and trends.
- **Admin Dashboard:** Provides administrative monitoring and control features.

8.1.2 Overview of Web Application

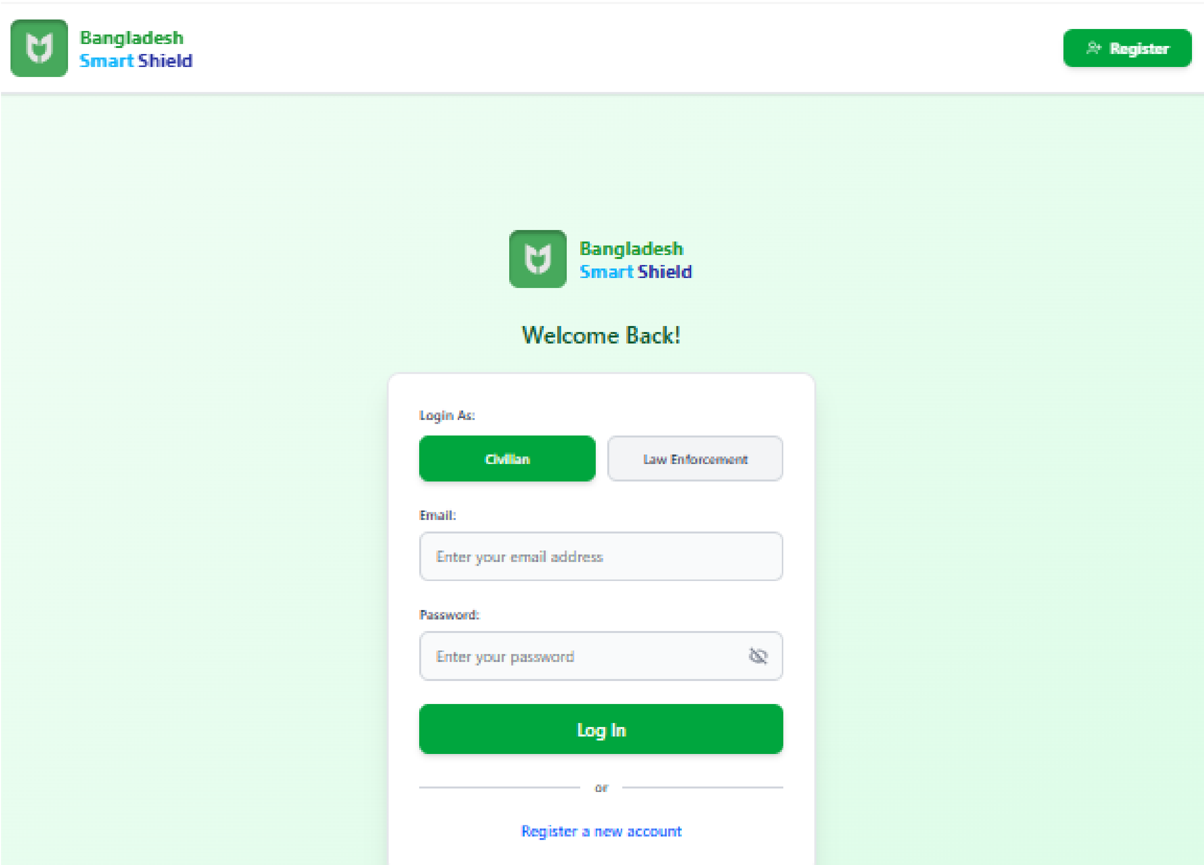

Figure 8.7: Login Page

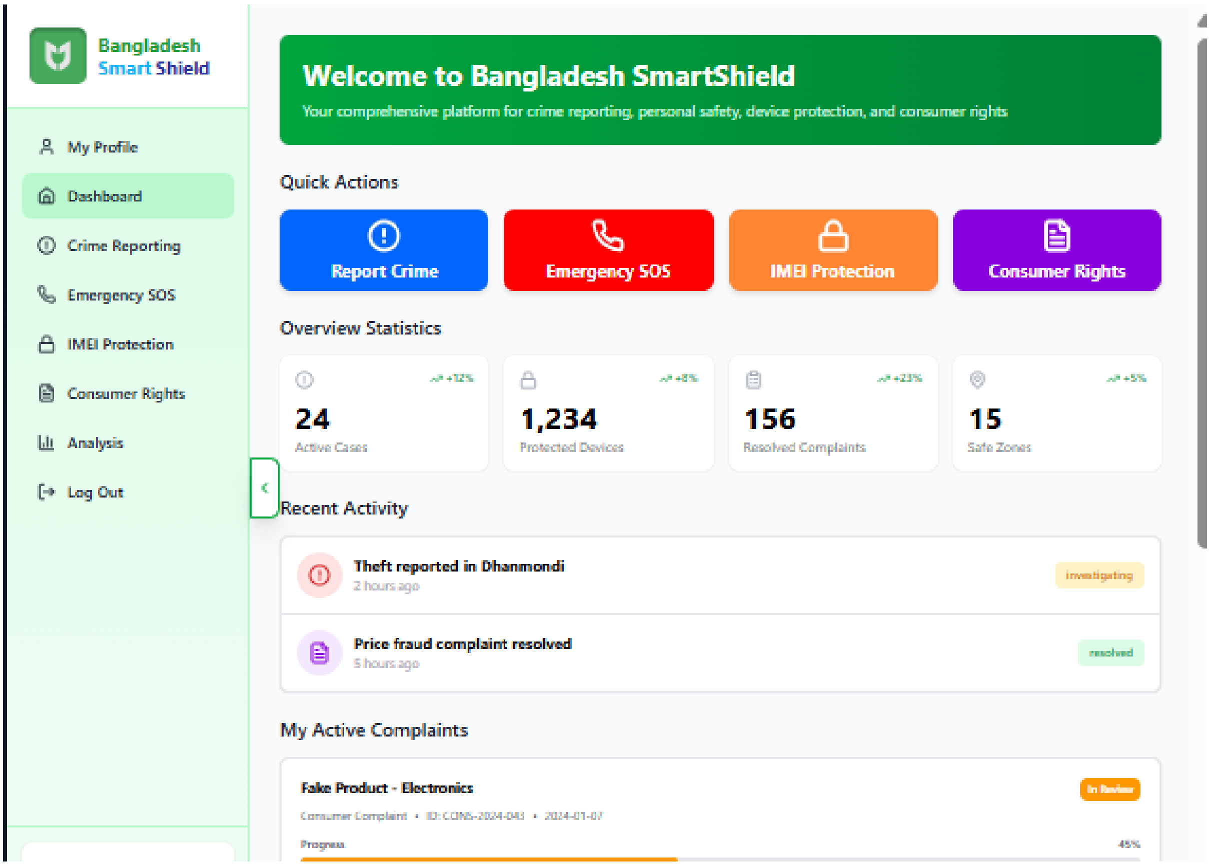

Figure 8.8: Dashboard Page

Figure 8.9: Crime Reporting Page

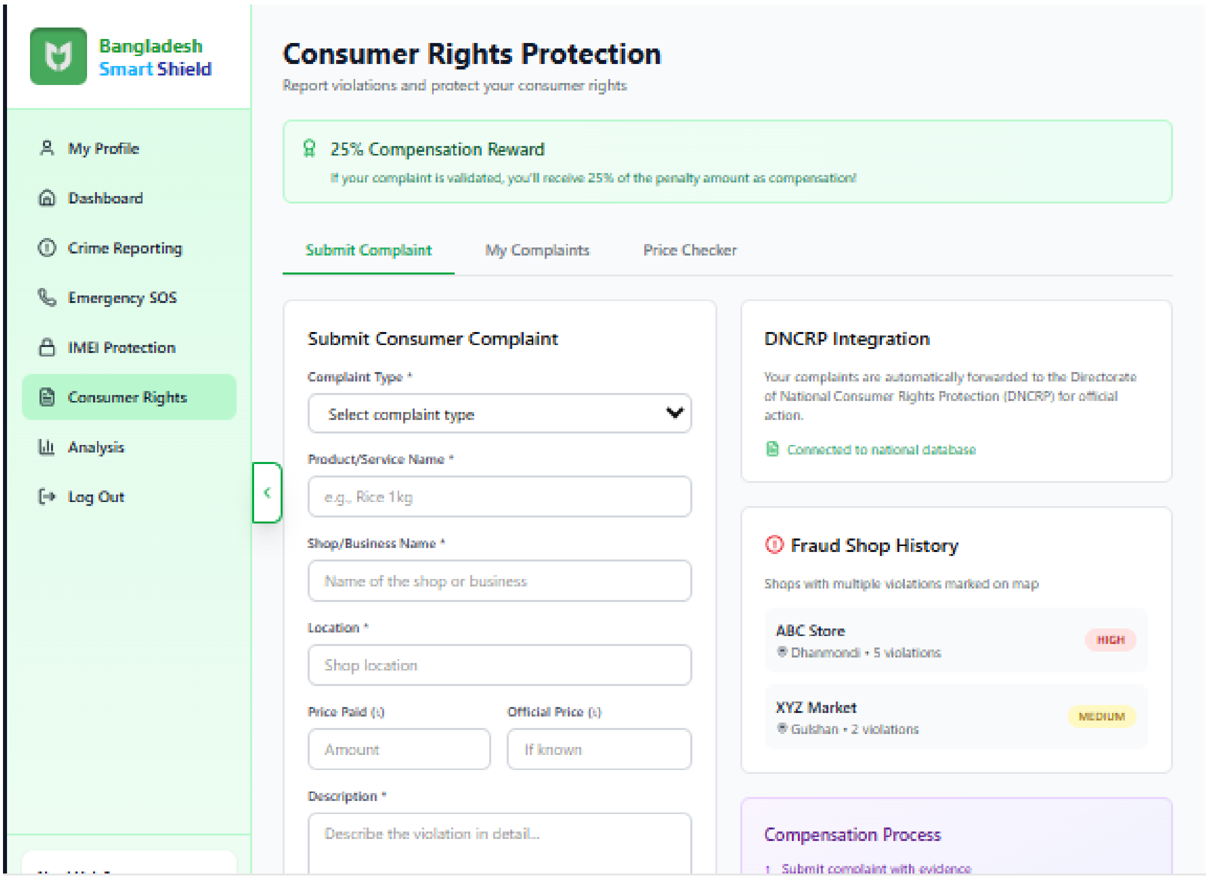

Figure 8.10: Consumer Rights Complaint Page

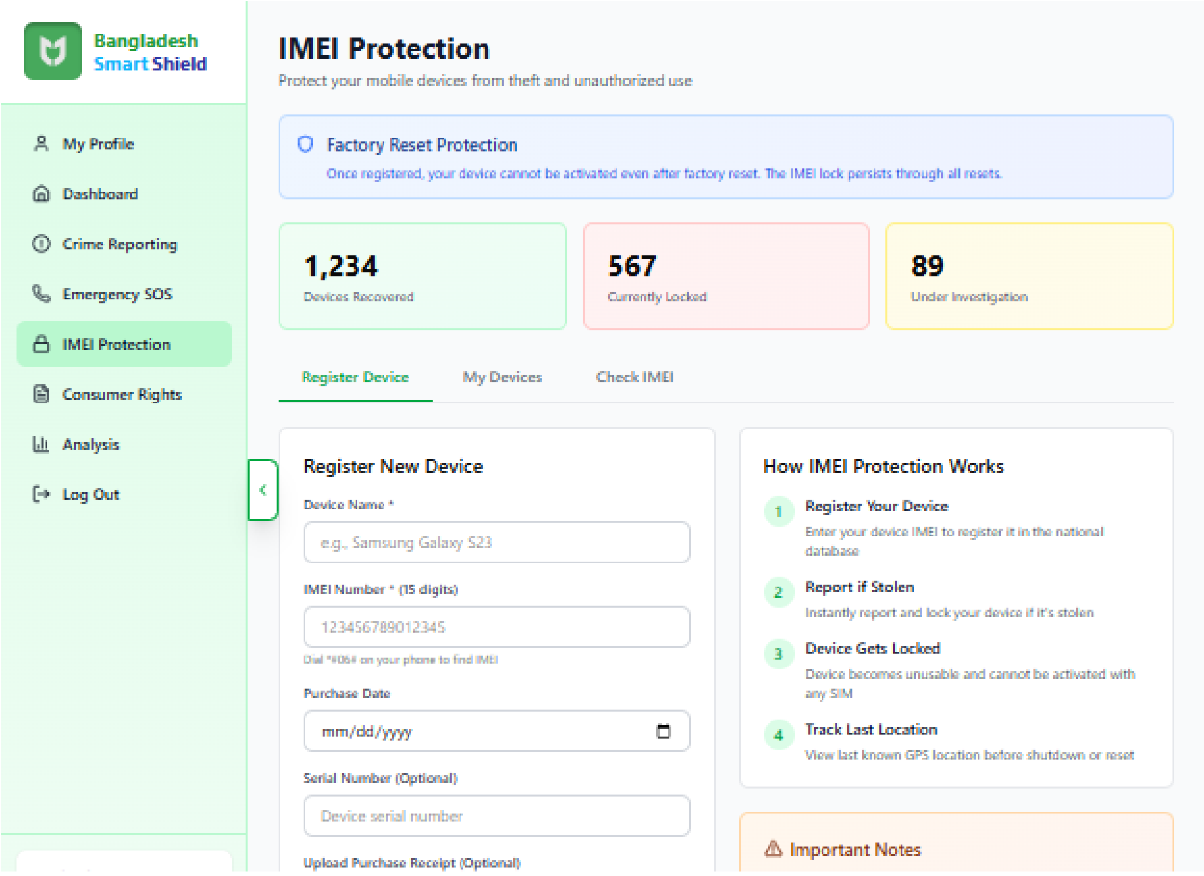

Figure 8.11: IMEI Protection Page

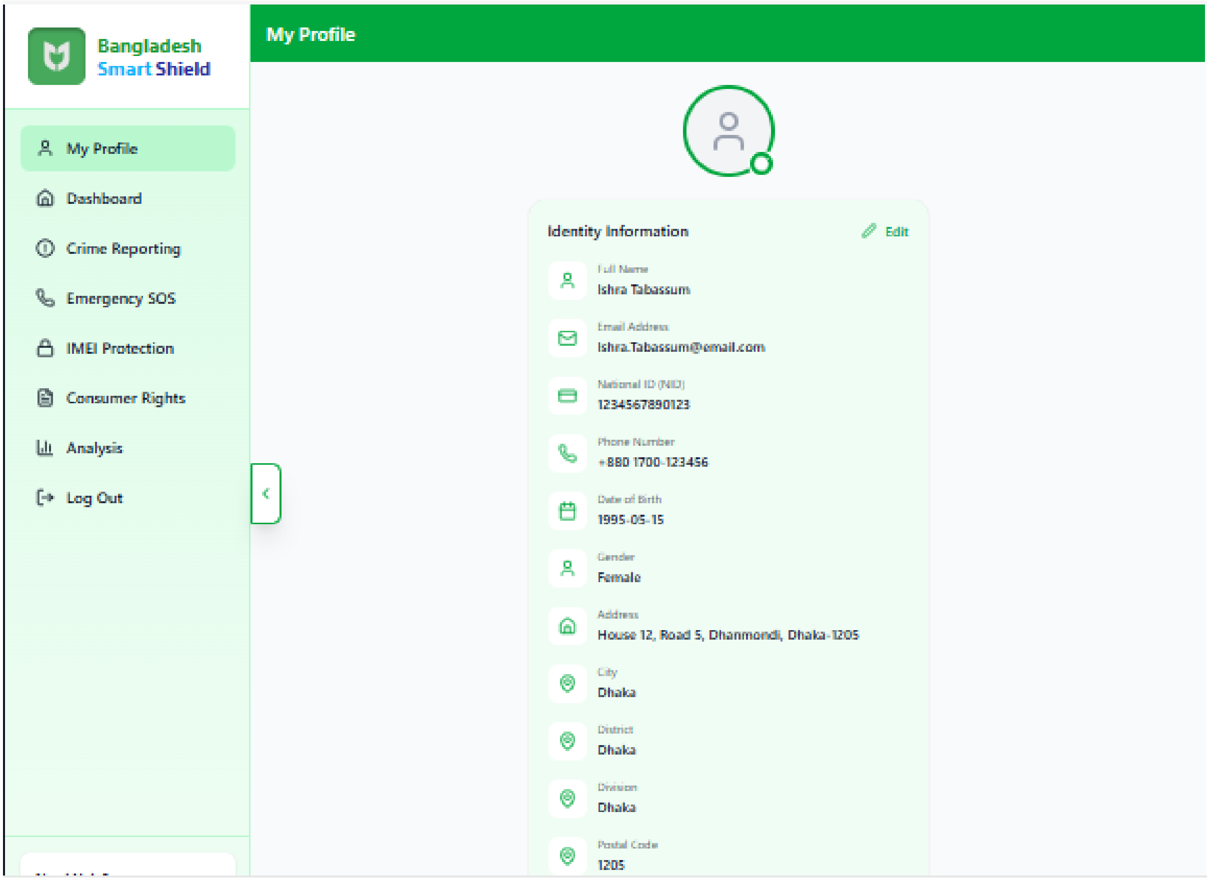

Figure 8.12: Profile Page

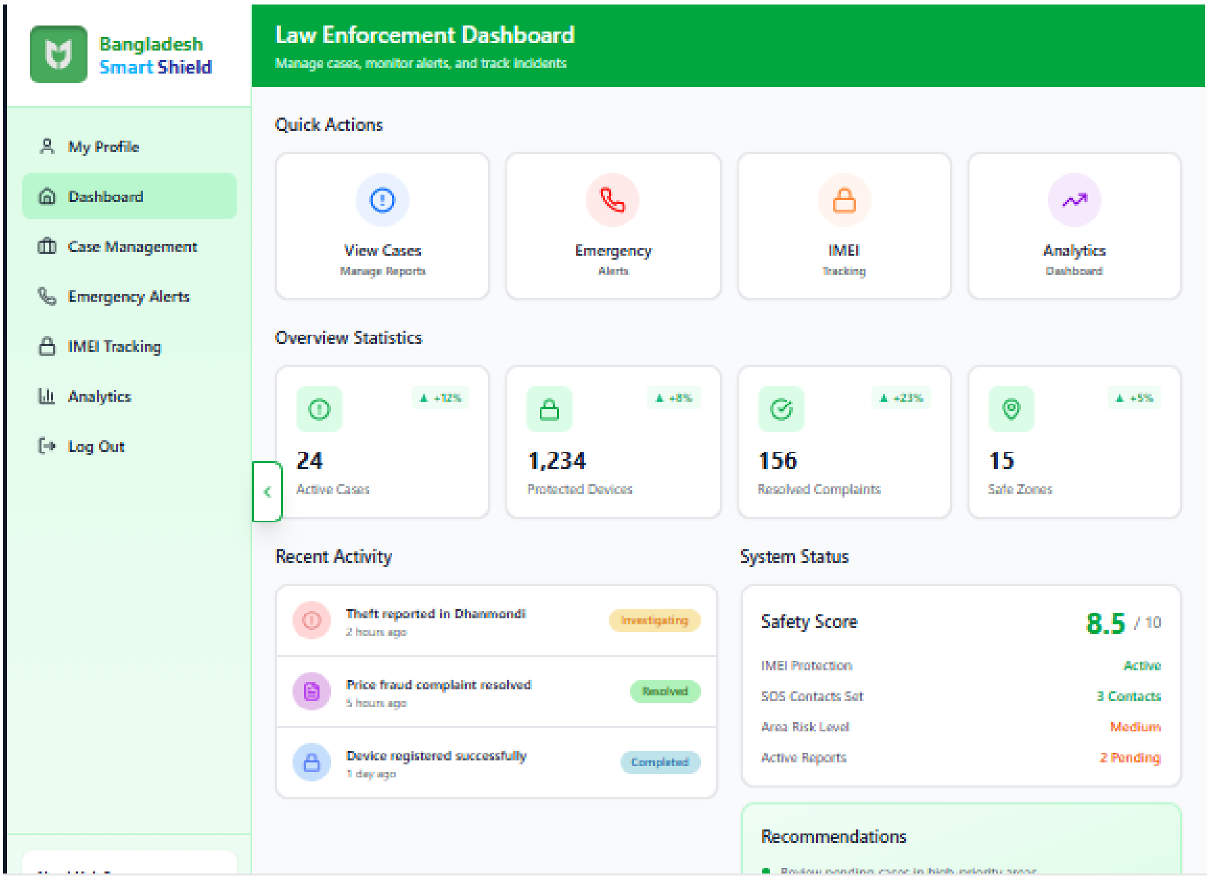

Figure 8.13: Law Enforcement Dashboard

8.2 UI Design Principles

The user interface of the Bangladesh SmartShield system follows Shneiderman's Eight Golden Rules of Interface Design to ensure usability, consistency, and user confidence across all system features.

1. **Strive for Consistency:** The system maintains consistent layouts, color schemes, icons, and navigation patterns across all modules to help users quickly understand and predict system behavior.

2. **Enable Frequent Users to Use Shortcuts:** Common actions such as SOS activation, crime reporting, and complaint tracking are designed to require minimal steps, allowing experienced users to operate the system efficiently.

3. **Offer Informative Feedback:** Every user action, such as form submission or emergency alert activation, provides immediate visual or textual feedback to confirm system response.

4. **Design Dialogs to Yield Closure:** Processes such as crime reporting, IMEI registration, and complaint submission are divided into clear steps with confirmation messages upon completion.

5. **Offer Simple Error Handling:** The system provides clear, understandable error messages and guidance to help users correct mistakes without frustration.

6. **Permit Easy Reversal of Actions:** Users are allowed to review, edit, or cancel actions (where applicable) before final submission to reduce anxiety and prevent accidental errors.

7. **Support Internal Locus of Control:** Users remain in control of system interactions, with predictable responses and transparent status updates for reported cases and complaints.

8. **Reduce Short-Term Memory Load:** The interface minimizes the need to remember information by displaying instructions, status updates, and contextual guidance directly on the screen.

Chapter 9

Conclusion

Bangladesh SmartShield consolidates national safety, digital protection, and consumer rights services into a single, unified, and verifiable system baseline. The platform integrates crime reporting, emergency response, IMEI-based mobile protection, and consumer complaint management to improve public trust, reduce response delays, and enhance transparency in digital governance.This Software Requirements Specification (SRS) document defines what the Bangladesh SmartShield system shall do through clearly stated functional requirements and specifies the quality attributes it must satisfy through non-functional requirements. Together, these provide a solid foundation for system design, implementation, validation, and acceptance.

What This SRS Establishes

- A coherent scope and stakeholder model with role-based capabilities for citizens, police, DNCRP officials, mobile operators, regulatory authorities, and system administrators.
- Verifiable performance, security, availability, and maintainability targets that guide engineering decisions and system evaluation.
- Traceable internal and external interfaces to ensure interoperability with emergency services, telecommunication systems, and consumer protection authorities, enabling future-ready integration.

Known Constraints

- IMEI blocking and stolen device enforcement depend on integration with mobile network operators and regulatory authorities.
- Real-time emergency alerts and location-based services rely on network availability and GPS accuracy.
- Data governance must comply with national privacy, cybersecurity, and data retention regulations.

Next Steps

1. Freeze requirements for the initial release, focusing on core features such as authentication, crime reporting, SOS emergency alerts, and consumer complaint submission.
2. Implement advanced modules including analytics dashboards, safety heatmaps, and factory-reset–resistant IMEI tracking iteratively with stakeholder feedback.
3. Perform performance, security, and usability verification against stated requirements, followed by pilot deployment, system stabilization, and formal handover to responsible authorities.

With this document serving as the contract of record, development can proceed incrementally, using defined acceptance criteria to validate that Bangladesh SmartShield delivers a secure, reliable, and citizen-centric national safety and protection platform.

Chapter 10

References

- IEEE Computer Society, *IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830)*.
- Bangladesh Police Official Website and Emergency Service (999) Information.
- Directorate of National Consumer Rights Protection (DNCRP), Government of Bangladesh.
- Bangladesh Telecommunication Regulatory Commission (BTRC) Publications and Guidelines.
- Software Engineering Textbook: Ian Sommerville, *Software Engineering*.
- Online resources related to digital governance, mobile security, and consumer protection.

Chapter 11

Appendices

11.1 Appendix A: Glossary

- **SRS:**Software Requirements Specification
- **SOS:**Emergency distress signal requesting immediate help
- **IMEI:**International Mobile Equipment Identity
- **NID:**National Identity Number
- **DNCRP:** Directorate of National Consumer Rights Protection
- **BTRC:**Bangladesh Telecommunication Regulatory Commission
- **API:**Application Programming Interface
- **GPS:**Global Positioning System
- **RBAC:**Role-Based Access Control

11.2 Appendix B: Additional Notes

This appendix may include future enhancements, system diagrams, or extended documentation as required during later phases of development.